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## PROFESSIONAL SUMMARY

- 10 years of experience in **Data Engineering, Data Pipeline Design, Development, and Implementation.**
- Expertise in **Hadoop** ecosystem components: **HDFS, YARN, MapReduce, Hive, Impala, Pig, Sqoop, HBase, Spark, Spark SQL, Kafka, Spark Streaming, Flume, Oozie, Zookeeper.**
- Experienced with **Hadoop/Hive** on **AWS (EMR and non-EMR-Hadoop in EC2).**
- Knowledge of **ETL** methods for data extraction, transformation, and loading in **Data Warehouse tools.**
- Develop data set processes for data modeling, data mining, data profiling and improving data reliability, efficiency, and quality.
- Extract Transform and Load data to Azure Data Storage using **Azure Cloud Data Factory, T-SQL, Spark SQL, Databricks, and Azure Synapse Analytics.**
- Working with **AWS/GCP cloud: GCP Cloud storage, Data-Proc, Data Flow, BigQuery, EMR, S3, Glacier, EC2 Instance.**
- Strong experience in scripting with **Python API, PySpark API, and Spark API** for data analysis.
- Installed and configured **Hive, Pig, Sqoop, Flume, and Oozie** on the **Hadoop cluster.**
- Implemented log producer in **Scala** for **Kafka** and **Zookeeper** based log collection platform.
- Experience in data manipulation with **R and Python.**
- Imported/exported data with **Sqoop** between **HDFS and Relational Databases**, loaded into partitioned **Hive tables.**
- Utilized **AWS utilities (EMR, S3, CloudWatch)** for running and monitoring **Hadoop and Spark jobs.**
- Developed **Spark code using Scala and Spark-SQL/Streaming** for faster data processing.
- Set up **Azure infrastructure** (storage accounts, integration runtime, service principal id, app registrations) for optimized utilization in Azure.
- Developed **MapReduce jobs** with **Pig, Hive, and Sqoop.**
- Developed and maintained web applications using **Django and Flask frameworks.**
- Experienced in code versioning tools like **Git** and **SVN**
- Created robust software applications with **Python**, focusing on clean code and scalability, meeting both business and technical requirements.
- Deployed Python applications to cloud platforms like **AWS and Azure**, using cloud services to ensure scalability and reliability.
- Built data pipelines using **Python frameworks** like **PySpark**, enabling efficient processing of large-scale data for various business needs.
- Specialized in leveraging advanced data querying and virtualization tools such as **Dremio, Presto, and Trino** to enhance data accessibility, performance, and scalability across diverse environments.
- Worked with data visualizing tools like **Tableau, Adobe Analytics, matplotlib**
- Extensive knowledge of **NoSQL databases: HBase, Cassandra, MongoDB.**
- Used **Azure PowerShell scripts** for data movement to **HDFS Blob storage.**
- Proficient in leveraging **SnowSQL** to optimize data migrations, automate ETL processes, and enhance query performance within Snowflake environments for improved data-driven decision-making.
- Worked with database solutions in **Azure SQL Warehouse, Azure SQL.**
- Analyzed **Hive** data with **Spark Data Frames API** over **Cloudera platform.**
- Developed **PySpark** scripts for data integration with third-party vendors.
- Experienced in working with **Oracle, SQL Server, Hive, and MongoDB** for ETL process validation.
- Extensive experience with **SQL and PL/SQL** for database development.

- Proficient with job workflow scheduling tools like **Oozie and Zookeeper**.
- Knowledgeable about **NoSQL** databases like **MongoDB, Azure, PostgreSQL, HBase, Cassandra**.
- Experience with **AWS services: EC2, S3, Auto Scaling, Redshift, DynamoDB, Route53**.

## AREAS OF EXPERTISE

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|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Big Data Technologies</b>         | Hadoop, Spark Core, Spark SQL, Hive, HBase, Parquet, Sqoop, Pig, Apache Kafka, Zookeeper, Flume, MapReduce Design Patterns, Data frames, Spark Streaming, Python, Hadoop Distribution (Cloudera, Horton Works), PySpark, Scala.                                                                                    |
| <b>Data Integration &amp; ETL</b>    | SQL, Azure Data Factory, Snowflake Modeling, Data Flow, Apache Kafka, Apache Flume, Apache Nifi, Apache Spark, Apache Beam, Apache Airflow, ELT/ETL Pipelines with Python, Snowflake Snow SQL, Spark Data Frames, Logic Apps, Azure Functions, CI/CD Frameworks (Jenkins), Data Ingestion, Azure Synapse Analytics |
| <b>Data Modeling &amp; Profiling</b> | Dimensional Modeling, Star Schema, Snowflake Schema, Slowly Changing Dimensions (SCD), Data Profiling, Data Quality Assessment, Entity-Relationship Diagrams (ERD), Normalization, Denormalization, Data Validation, Data Cleansing.                                                                               |
| <b>Version Control</b>               | GIT, GitHub, Jenkins, GitHub Actions, Terraform                                                                                                                                                                                                                                                                    |
| <b>Databases</b>                     | Azure SQL DB, Azure Synapse. MS Excel, MS Access, Oracle 11g/12c, Cosmos DB, mango Db                                                                                                                                                                                                                              |
| <b>Data Storage &amp; Retrieval</b>  | Azure Storage, Azure SQL DB, Blob Storage, MySQL, Cassandra, Snowflake, Azure Data Lake Storage (adls), Gen2, Azure Data Bricks, Data Lakes, HDFS, SQL                                                                                                                                                             |
| <b>Data Analysis &amp; Reporting</b> | Power BI, Tableau, Excel Sheets, Flat Files, CSV Files, Text files, SQL Reporting Services.                                                                                                                                                                                                                        |

## EDUCATION:

- Bachelor's from Anna University

## CERTIFICATION:

- **Azure Data Fundamentals -DP 900**
- **Azure Data Engineer Associate - DP 203**
- **Databricks Data Engineer Associate**

## WORK EXPERIENCE:

**Client: Molina Health Care, Long Beach, CA**

**Feb 2023 to Present**

**Role: Sr. Data Engineer**

- Implemented end-to-end data pipelines using Azure Data Factory to extract, transform, and load (ETL) data from diverse sources into Snowflake.
- Designed and implemented data processing workflows using Azure Databricks, leveraging Spark for large-scale data transformations.
- Built scalable and optimized Snowflake schemas, tables, and views to support complex analytics queries and reporting requirements.
- Built scalable and optimized **Snowflake schemas**, tables, and views to support complex analytics queries and reporting requirements.
- **Performed data profiling** to assess data quality, identify anomalies, and ensure data integrity across multiple datasets.

- Designed and implemented **data models** (star schema, snowflake schema) to support business intelligence and reporting needs.
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- Developed data ingestion pipelines using Azure Event Hubs and Azure Functions to enable real-time data streaming into Snowflake.
- Leveraged Azure Data Lake Storage as a data lake for storing raw and processed data, implementing data partitioning and data retention strategies.
- Utilized Azure Blob Storage for efficient storage and retrieval of data files, implementing compression and encryption techniques to optimize storage costs and data security.
- Integrated Azure Data Factory with Azure Logic Apps for orchestrating complex data workflows and triggering actions based on specific events.
- Implemented data governance practices and data quality checks using Azure Data Factory and Snowflake, ensuring data accuracy and consistency.
- Implemented data replication and synchronization strategies between Snowflake and other data platforms using Azure Data Factory and Change Data Capture techniques.
- Developed and deployed Azure Functions for data preprocessing, data enrichment, and data validation tasks in data pipelines.
- Implemented advanced analytics and machine learning workflows using Azure Machine Learning and Snowflake, enabling predictive analytics and data-driven insights.
- Designed and implemented data archiving and data retention strategies using Azure Blob Storage and Snowflake's Time Travel feature.
- Developed custom monitoring and alerting solutions using Azure Monitor and Snowflake Query Performance Monitoring (QPM) for proactive identification and resolution of performance issues.
- Integrated Snowflake with Power BI and Azure Analysis Services for creating interactive dashboards and reports, enabling self-service analytics for business users.
- Optimized data pipelines and Spark jobs in Azure Databricks for improved performance, including tuning of Spark configurations, caching, and leveraging data partitioning techniques.
- Implemented data cataloging and data lineage solutions using tools like Azure Purview and Apache Atlas to provide a comprehensive understanding of data assets and their relationships.
- Collaborated with cross-functional teams, including data scientists, data analysts, and business stakeholders to understand data requirements and deliver scalable and reliable data solutions.

**Environment:** Azure SQL Database, Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure Kubernetes, Azure DevOps, ADLS Gen2, Event Hubs, PolyBase, PySpark, HDFS, MapReduce, Spark, Spark Streaming, Kafka, Hive, ORC, Parquet, Text File, Compression Techniques, Dimensional Modeling, Slowly Changing Dimensions (SCD), Star Schema, Snowflake Schema, SQL Server, Oracle, Vertica, Teradata.

**Client:** Northern Trust, Chicago  
**Role:** Azure Data Engineer

**Sep 2020 to Feb 2023**

- Leveraged PySpark to design and implement ETL pipelines on Azure Databricks, directly addressing complex business challenges and streamlining data transformation processes.
- Developed and managed data validation workflows in Spark SQL within Databricks environments, ensuring data accuracy and consistency across multiple datasets.
- **Performed data profiling** to identify data quality issues and ensure data consistency across multiple sources.
- Designed and implemented **data models** to support business intelligence and reporting needs, including star and snowflake schemas.
- Orchestrated data workflows with dependency management on Azure Databricks, optimizing big data processing for enhanced performance and reliability.
- Automated job scheduling in Azure Data Factory (ADF) use a variety of triggers such as Events, Schedules, and Tumbling, ensuring timely data processing and job execution.
- Built ADF pipelines with Linked Services, Datasets, and Pipelines to seamlessly extract, transform, and

load data from sources, including Azure SQL, Blob storage, Snowflake, and Azure Synapse Analytics.

- Designed secure ADF pipelines that dynamically retrieve secrets from Azure Key Vault, implementing scalable and secure parameter management during pipeline execution.
- Delivered essential datasets for data science model creation on Azure Databricks, supporting machine learning initiatives with high-quality, pre-processed data.
- Enhanced SQL database query performance by applying Azure best practices, achieving multiple times faster execution for high-demand business applications.
- Applied advanced SQL querying techniques to validate data and construct a reliable data warehouse in Azure Databricks, improving data integrity and accessibility.
- Configured Azure Databricks workflows for continuous job execution, enabling efficient and uninterrupted data management in a big data environment.
- Designed and managed ADF Pipelines using Linked Services, Datasets, and Dynamic Parameterization to streamline data integration.
- Developed APIs in Snowflake and ADF to support data exchange and workflow automation.
- Implemented data governance policies in Snowflake to enforce security, access control, and compliance.
- Automated batch processing and PowerShell scripting to optimize ETL processes.
- Migrated large-scale on-premises databases to Snowflake, improving storage efficiency and data accessibility.

**Environment:** Azure Databricks, Azure Data Factory, Azure Logic Apps, Functional Apps, MySQL, Oracle, HDFS, MapReduce, Apache Spark, Apache Hive, Python, Scala, PySpark, PostgreSQL, .NET, Teradata, Shell Scripting, GIT, JIRA, Jenkins, Apache Kafka, Power BI, Azure Synapse Analytics, Polybase, Azure Data Lake Storage, Azure SQL Database, Azure DevOps.

**Client:** US Bank, CA.

**Dec 2018 to Sep 2020**

**Role:** Data Engineer

- Wrote and optimized SQL queries for data extraction, transformation, and loading (ETL) processes, ensuring accurate and efficient data processing.
- Designed and implemented normalized and denormalized data schemas, enhancing data integrity and retrieval efficiency in various projects.
- Developed and automated ETL pipelines using various tools, streamlining data integration and transformation workflows.
- Performed data profiling to assess data quality and ensure consistency across datasets.
- Designed and managed data models to align with business requirements, optimizing data storage and retrieval strategies.
- Utilized AWS S3 for storing, managing, and retrieving data, ensuring scalable and secure data storage solutions.
- Leveraged AWS Lambda for serverless data processing tasks, enabling efficient and cost-effective computation and automation.
- Managed and optimized databases with Amazon RDS, delivering reliable and scalable cloud-based database solutions.
- Applied Informatica for data integration, transformation, and loading, improving the efficiency of data workflows.
- Utilized Snowflake for cloud-based data warehousing, optimizing data storage, and querying performance.
- Employed Python for scripting and automating data tasks, improving efficiency, and reducing manual data handling.
- Managed code repositories and tracked changes using Git, ensuring effective version control and collaboration on data engineering projects.
- Leveraged Apache Spark for large-scale data processing and analytics, enhancing the handling and analysis of big data.
- Applied networking knowledge in AWS, including configuring Virtual Private Cloud (VPC), subnets, and security groups to ensure secure and efficient cloud infrastructure.

- Participated in Agile teams and Scrum methodologies, using JIRA to track project progress and facilitate effective team collaboration.
- Developed and maintained data pipelines for ETL processes, automating data integration and ensuring seamless data flow.
- Designed and managed data models to align with business requirements, optimizing data storage and retrieval strategies.
- Implemented automated data processing workflows using Python and ETL tools, increasing operational efficiency and accuracy.
- Supported the deployment and management of cloud-based data solutions using AWS services, ensuring scalable and secure infrastructure.
- Integrated data from various sources using Informatica, improving data quality and accessibility for analysis.
- Applied Agile and Scrum practices to deliver data engineering solutions efficiently, ensuring timely completion of project milestones.
- Utilized Git for version control and collaboration, ensuring consistent and effective development practices across the team.

**Environment:** Azure SQL Database, Azure Data Factory (ADF) V2, Azure Key Vault, Azure Blob Storage, PySpark, Spark SQL, RESTful API, Kafka, Kinesis APIs, Python libraries (Pandas, NumPy, Scikit-learn), Pub/Sub messaging systems, Hadoop, Spark, Flink, CDC pipelines, Flask, Jenkins, AJAX, JSON, Git, GitHub, Agile, JIRA.

**Client:** Prolifics, INDIA  
**Role:** Business Analyst

**July 2013 to Oct 2016**

- Collaborated with clients to gather and analyze requirements, working closely with teams to create detailed designs and coordinating with client teams to confirm and implement design changes for alignment with project goals.
- Designed and implemented Hive tables for data storage and retrieval, efficiently managing data inserts and executing MapReduce-based queries to support large-scale analytics.
- Configured and scheduled automated SSIS jobs for data import, transformation, and cleansing, optimizing data management and operational workflows across various environments.
- Converted complex SQL queries to Spark transformations, leveraging Spark RDDs and Scala for improved performance and scalability in data processing tasks.
- Implemented Kafka for real-time data ingestion into Hadoop, creating Oozie jobs to manage and automate data workflows for streamlined processing.
- Extracted and loaded log data from web servers into HDFS, setting up infrastructure for further data analysis and insights extraction.
- Created and managed Hadoop job flows while troubleshooting and optimizing workflows by reviewing log files to ensure seamless data operations.
- Designed and structured HBase tables with specific columns to efficiently store and query user event data, ensuring rapid data retrieval and storage efficiency.
- Migrated databases from traditional data warehouses to Spark clusters, facilitating the transition to more flexible, high-performance data architectures.
- Queried data in HBase using Hive and Impala, enabling fast and efficient access to large data volumes for analytics and reporting needs.
- Developed PL/SQL stored procedures, functions, triggers, and views using indexing, aggregation, and materialized views to optimize query performance across systems.
- Configured and deployed Oozie workflows on a Hadoop cluster, running Spark and Pig jobs parallel to enhance data processing efficiency.

- Implemented dynamic partitioning and bucketing in Hive, optimizing data organization and ensuring quick access to large datasets.
- Built ETL pipelines using Spark and Hive, ingesting data from multiple sources and transforming it for downstream consumption in analytics and reporting.
- Led a successful data migration from legacy systems to Snowflake, managing the seamless transfer of large data volumes and optimizing storage for improved accessibility.
- **Environment:** Power BI, SQL Server, Python, Oracle, SSIS, Snowflake.